MOI

MADINAH OPERATIONS INTELLIGENCE

- Overview
- Operations
- AI Insights
- Project Info

PORTFOLIO PROJECT
Synthetic data
Inspired by internship experience

Madinah Operations Intelligence

From municipal reports to proactive decisions

Date Range: 05/10/25 – 05/04/26

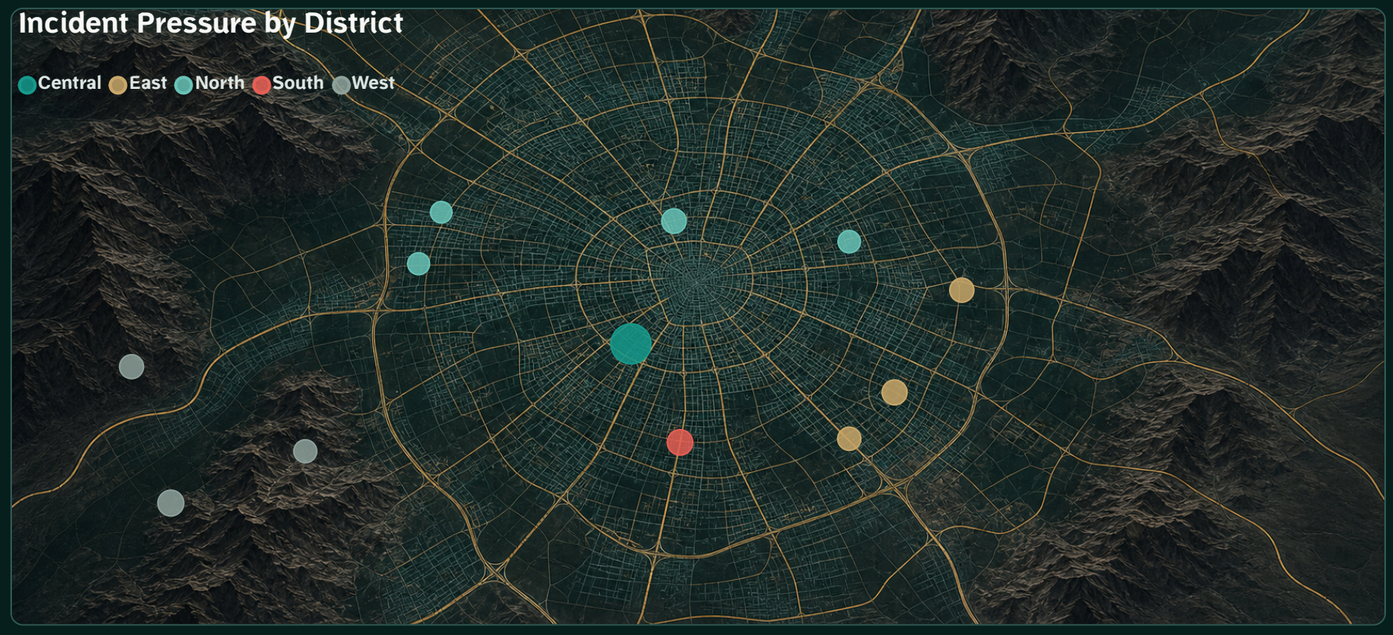
District: All

Category: All

Shift: All

30K	73.5%	28.3	8.4%
Incoming Reports	SLA Compliance	Resolution Hours	Repeat Rate

Incident Pressure by District



Operational Insight

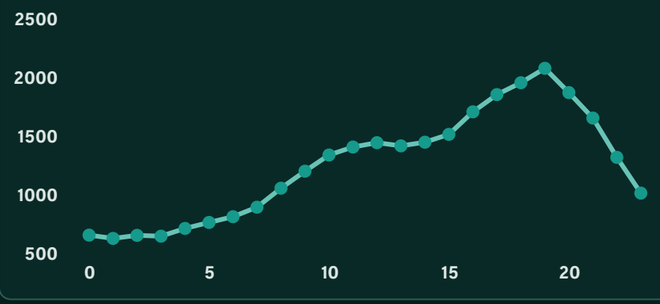
Finding
35.7% of reports in All Districts occur during the evening period.

Repeat rate: 8.4%

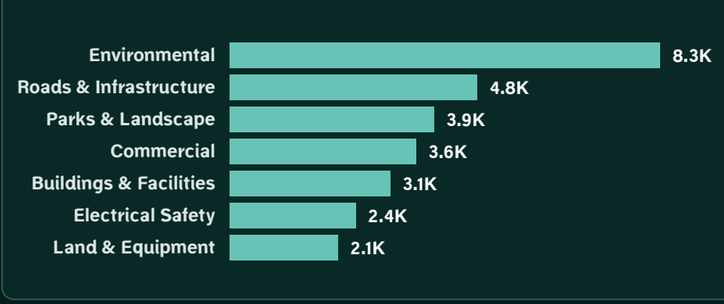
Suggested action
Review demand by shift and repeat-report locations.
Validate capacity allocation before staffing decisions.

AI-assisted

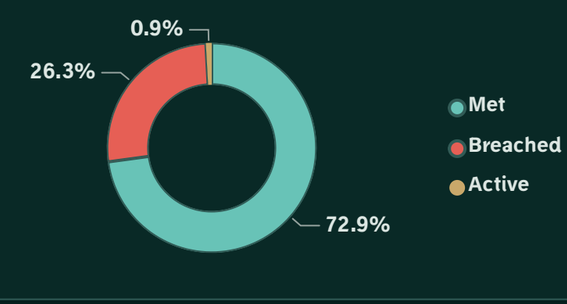
Incident Demand by Hour



Reports by Service Category



SLA Status



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Operations & SLA Performance

Capacity, workload and service-level monitoring

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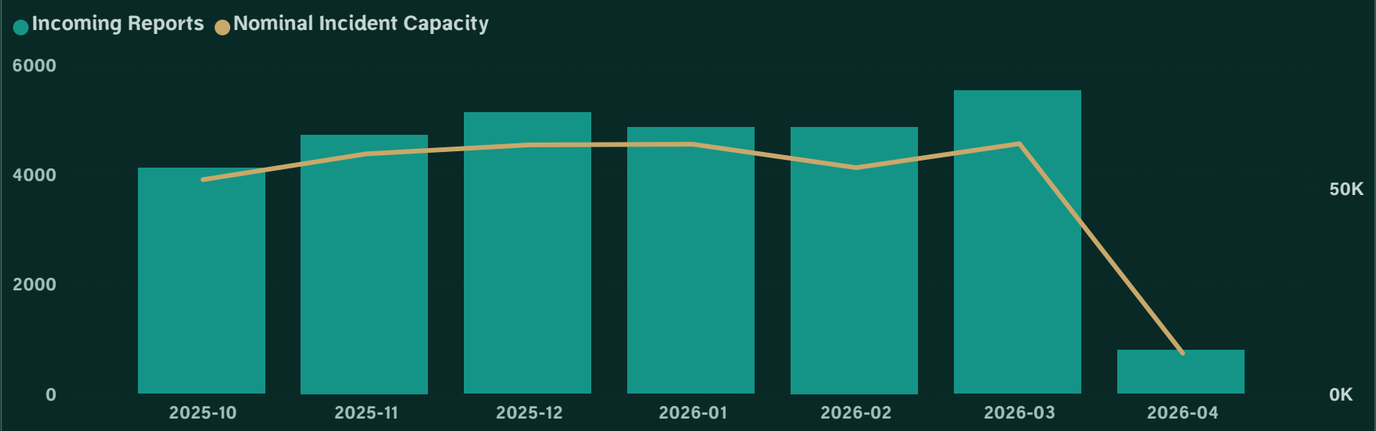
District: All

Shift: All

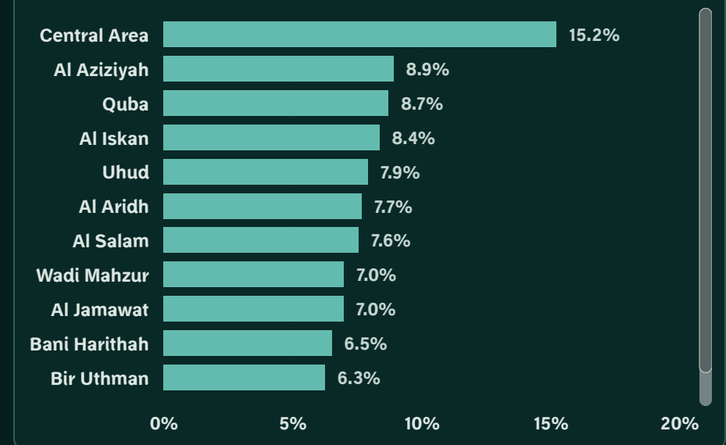
357K	8.4%	97.3%	267
Nominal Incident Capacity	Demand / Nominal Capacity	Team Availability Rate	Active Reports

Demand vs Nominal Capacity

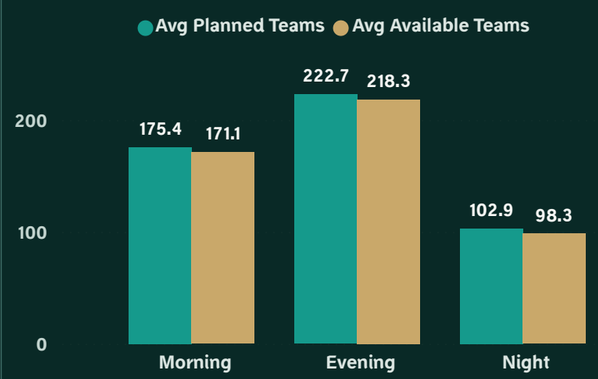
Monthly totals · Apr 2026 includes data through 5 Apr



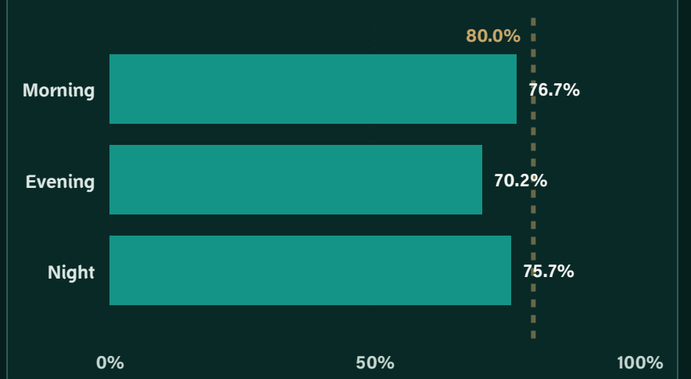
Demand / Nominal Capacity by District



Average Daily Team Allocations by Shift



SLA Compliance by Shift



District Operational Priorities

District	Demand / ...	SLA	Active
Uhud	7.9%	77.8%	21
Quba	8.7%	77.6%	22
Central Area	15.2%	59.3%	63
Al Iskan	8.4%	78.7%	23
Al Aziziyah	8.9%	78.1%	29
Al Aridh	7.7%	77.3%	12
Total	10.0%	71.2%	170

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AI-Assisted Operational Insights

Risk detection, demand signals and explainable recommendations

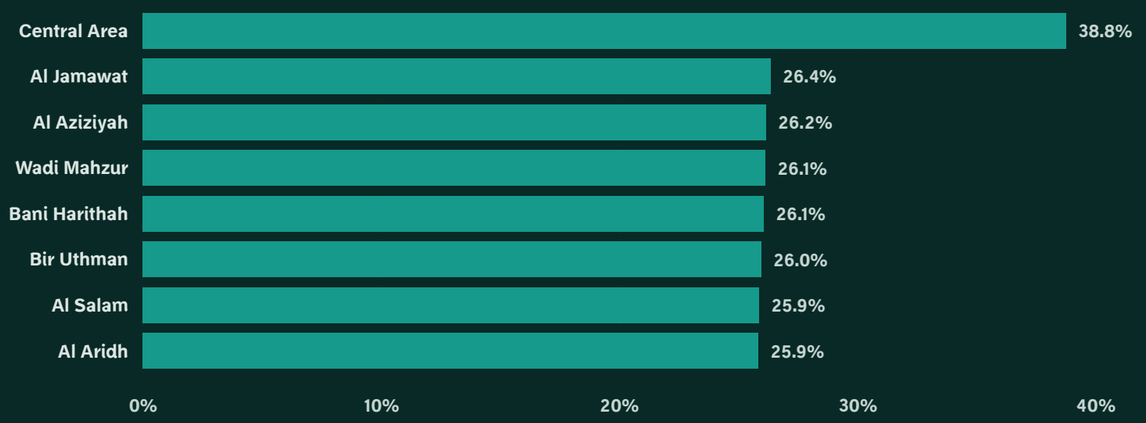
Date Range: 05/10/25 – 05/04/26

District: All

Shift: All

28.8%	3371	39	85.3%
Average Predicted Risk	High Risk Reports	Active High Risk	Model Recall

Predicted SLA Risk by District



Explainable Recommendation

AI Recommendation

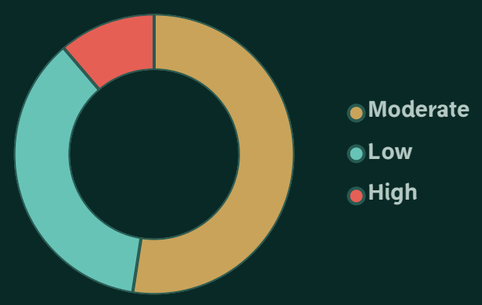
Priority district
Central Area

Predicted risk: 38.8%
Active high-risk: 32
Demand / nominal capacity: 15.2%

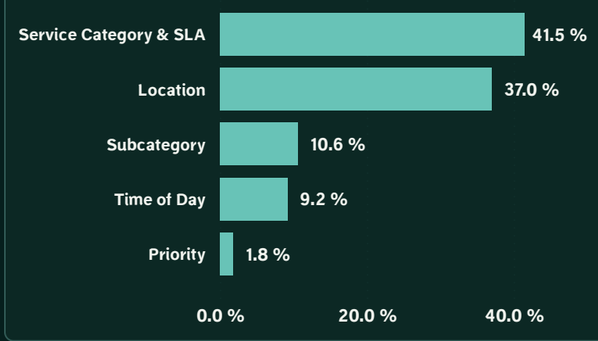
Review active high-risk reports and repeat locations. Synthetic capacity; validate allocation before staffing decisions.

AI-assisted; human review required

Predicted Risk Distribution



Model Drivers



High-Risk Priority Queue

Incident ID	District	Risk	Priority	Shift
SYN-029957	Central Area	59.7 %	High	Evening
SYN-029999	Central Area	53.8 %	High	Night
SYN-029818	Central Area	50.5 %	High	Evening
SYN-029621	Central Area	50.4 %	Low	Evening
SYN-029628	Central Area	50.4 %	Low	Evening
SYN-029973	Central Area	50.4 %	Low	Evening
SYN-029812	Central Area	49.9 %	Low	Evening
SYN-029645	Central Area	49.2 %	Medium	Evening
SYN-029813	Central Area	49.2 %	Medium	Evening
SYN-029478	Central Area	48.7 %	Medium	Evening
SYN-029489	Central Area	48.7 %	Medium	Evening

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Project Overview & Methodology

Business context, data architecture and AI approach

Project Summary

Madinah Operations Intelligence is an end-to-end BI and predictive analytics portfolio project inspired by internship experience at the Municipal Operations Monitoring Center. It transforms synthetic municipal report data into operational monitoring, capacity analysis, SLA risk prediction and explainable decision support.

SYNTHETIC DATA ▪ PRIVACY-SAFE ▪ PORTFOLIO PROTOTYPE

Project Facts

INTERNSHIP PERIOD

05 Oct 2025 – 05 Apr 2026

DOMAIN

Municipal Operations Intelligence

DATASET

30,000 synthetic municipal reports

CORE TOOLS

Power BI ▪ Python

DAX ▪ Power Query

FOCUS

BI ▪ Operations

Predictive AI

From Municipal Data to Proactive Decisions

01 SYNTHETIC DATA

02 POWER QUERY & MODEL

03 BI & DAX ANALYTICS

04 AI RISK SCORING

05 DECISION SUPPORT

Business Problem

- High report concentration near the Central Area
- Uneven demand / nominal capacity by district
- SLA breaches and repeat-report pressure
- Need for proactive prioritization

BUSINESS OUTCOME

Faster, evidence-based operational decisions

AI Methodology

Target: SLA breach before resolution

Models: sklearn Logistic Regression + AdaBoost

Selected: AdaBoost Decision Stumps

Time cohorts; labels known before each cutoff

Staffing excluded; feature assumptions remain

AI-ASSISTED ▪ HUMAN REVIEW REQUIRED

Model Performance & Governance

TEST ROC-AUC	68.7%
MODEL RECALL	85.3%
DECISION THRESHOLD	24%

Synthetic data ▪ Six-month history

Retrospective prototype ▪ Censoring limits apply

Predictions support—not replace—human decisions